

Benjamin Mellin

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EDUCATION

University of Florida

Bachelor of Science in Materials Science and Engineering, Cum Laude
Associate's of Arts

Gainesville, FL

May 2026

Dec. 2023

EXPERIENCE

Lead Materials Engineer

Aug. 2025 – May 2026

Honeywell Aerospace | Atmospheric Re-Entry Control

Clearwater, FL

- Led the physical prototype sub-team and owned the full materials selection workstream within a 7-person team, developing a custom S.M.A.R.T. (Simulation and Machine-learning Assisted Redesign and Testing) methodology that replaced traditional trial-and-error with an automated, data-driven design loop using Kriging models and ANSYS simulation to optimize a housing for atmospheric demise.
- Engineered deployable flaps to increase surface heating during re-entry, validating a design at **548 MW/m² heat flux** and **1,443 K average temperature** against Final Design Review targets of 400 MW/m² and 1,200 K, clearing every Technical Performance Measure.
- Documented and communicated team progress through a weekly public blog, maintaining technical transparency with Honeywell Aerospace industry sponsors throughout the design cycle.

Team Lead

Mar. 2025 – Apr. 2025

1075 Steel Heat Treatment Design

Gainesville, FL

- Directed a full heat treatment design process for 1075 steel, including literature review, phase diagram analysis, and TTT diagram interpretation to select an optimal heat-and-quench-plus-temper processing route.
- Martensitically transformed and tempered the steel to achieve target hardness, achieving a **60% increase in Young's Modulus**; presented results as a research poster.

Materials Selection Lead

Fall 2025

Superalloy Design for Hydrogen Turbine Blades | EMA4714 Materials Selection & Design

Gainesville, FL

- Selected Inconel 718 via a custom material index balancing fracture toughness, yield strength, density, and cost to resist oxygen-assisted intergranular cracking and hydrogen embrittlement in a hydrogen-fueled gas turbine blade, meeting targets of $>80 \text{ MPa}\sqrt{m}$ fracture toughness and $>100 \text{ ksi}$ yield strength.
- Demonstrated that EI-MFC post-processing increased the strengthening γ' phase fraction from **57.1% to 70.7%**, directly reducing hydrogen diffusivity at grain boundaries.
- Built the full cost model for a 100-unit production batch via vacuum investment casting, estimating **\$3,250 per blade** and identifying direct labor and R&D as dominant cost drivers.

Failure Analysis Engineer

Fall 2025

Duplex Stainless Steel Impeller Failure Analysis | Materials Failure Analysis

Gainesville, FL

- Performed SEM and EDS analysis on a super duplex stainless steel submersible pump impeller that failed after one year in a phosphate mining slurry line for industry sponsor Townley Foundry & Machine.
- Quantified porosity via SEM/ImageJ void analysis, finding **174 voids in the failed sample versus 28 in the reference**, and traced the failure to a mixed brittle-ductile fracture driven by surface stress concentrators.
- Delivered casting and installation recommendations — including argon-purge degassing and improved shaft alignment — to reduce recurrence in future production.

Team Lead

Aug. 2024 – Dec. 2024

Engineering Innovation

Gainesville, FL

- Led a cross-functional team through the full product development lifecycle, from need analysis and market research through brainstorming, prototyping, and investor-style pitch.
- Conceptualized and pitched MindMate, a wearable AI assistant for elderly task management; developed a comprehensive innovation playbook and presented to a faculty jury in the UF innovation department.

Volunteer Foster

May 2025 – Present

Low Rider Rescue

Wesley Chapel, FL

- Provide structured foster care for dogs, tracking health and behavioral progress and communicating observations to rescue coordinators and prospective adopters.

SKILLS

Software & Data Tools: ANSYS, SolidWorks, MATLAB, Python, GRANTA (material identification, sustainability, cost analysis), Excel

Engineering: Thermal-Structural Analysis, Material Characterization, Failure Analysis, Composite Production, Product Development, DLP & FDM 3D Printing, Arduino, Deconstruction

Testing Techniques: Tensile (Instron UTS), Charpy Impact, 3-Point Bending, Scanning Electron Microscopy, X-ray Diffraction, Fourier Transform Infrared (FTIR) Spectroscopy, Degradation Testing

Communication: Technical Writing (SOPs, research papers, failure analysis reports), Public Speaking, Poster and PowerPoint Presentation, Group Leadership, Literature Review, Six Sigma Quality White Belt